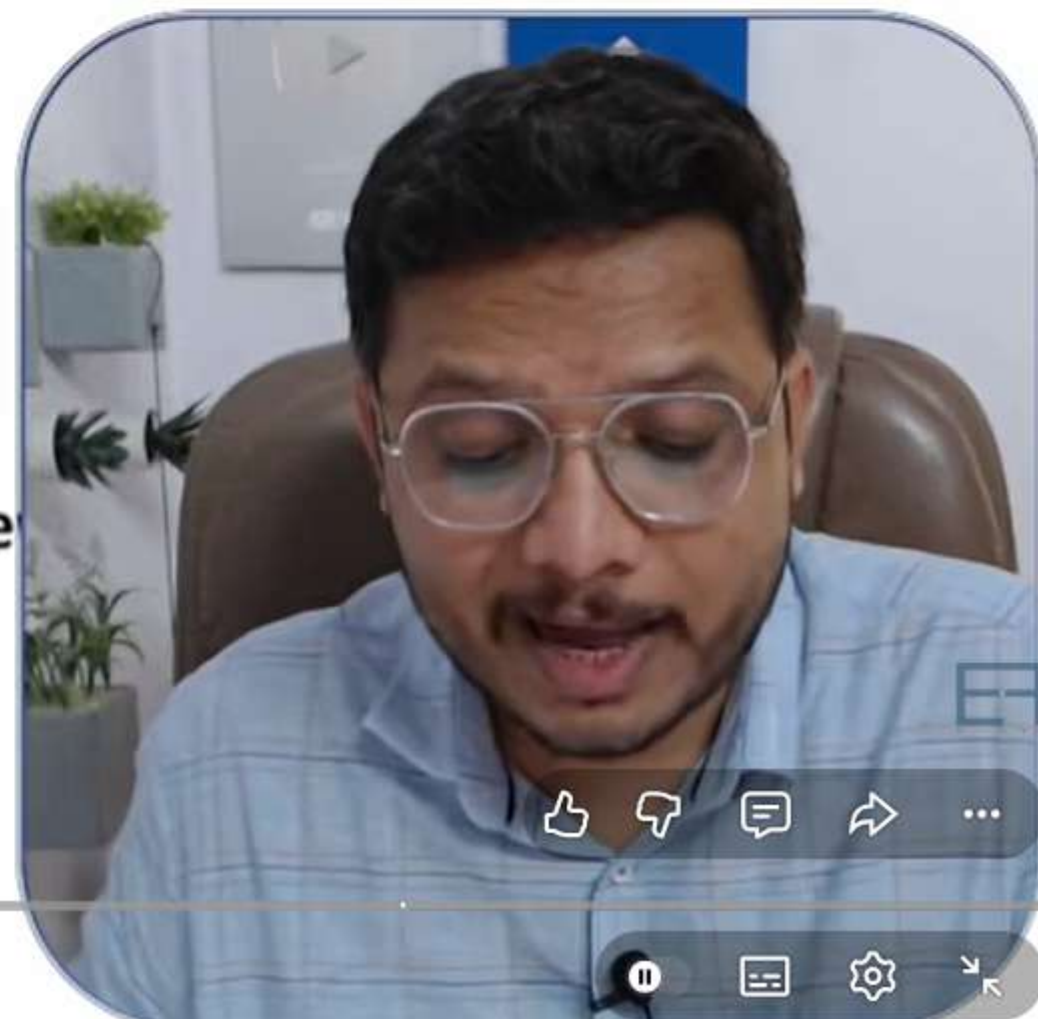
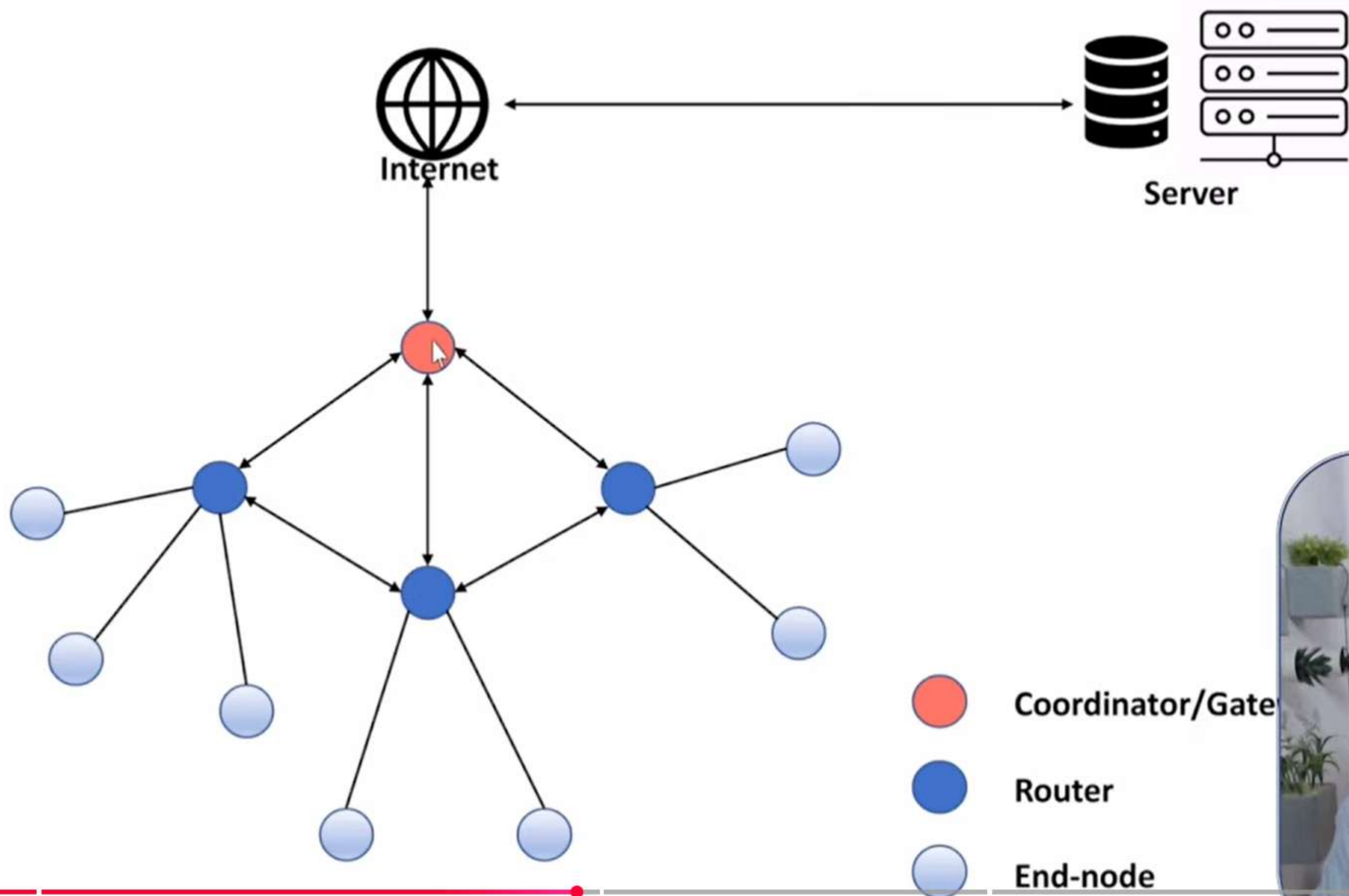


Wireless Sensor Networks





Basics of Wireless Sensor Networks

- A Wireless Sensor Network (WSN) comprises of distributed devices with sensors that are used to monitor the environmental and physical conditions.
- A WSN consists of a number of end nodes and routers and a coordinator.
- End nodes have several sensors attached to them. End nodes can also act as routers.
- Routers are responsible for routing the data packets from end nodes to the coordinator.
- The coordinator collects the data from all the nodes.
- The Coordinator also acts as a gateway that connects the WSN to the Internet.



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Basics of Wireless Sensor Networks - WSN >



Characteristics of Wireless Sensor Networks

- Wireless Protocols (ZigBee, Bluetooth, Wi-Fi etc)
- Range (10m to 100m)
- Speed (100KBps to Few MBps)
- Operating Frequency (ISM Bands, 2.4GHz in majority devices)
- Low Power Operating nodes.
- WSNs are self-organizing networks.
- Robust structure. (Even if some nodes are failing, it does autoconf nodes)



Examples of Wireless Sensor Networks

- Weather monitoring systems use WSNs in which the nodes collect temperature, humidity and other data, which is aggregated and analyzed.
- Indoor air quality monitoring systems use WSNs to collect data on the indoor air quality and concentration of various gases.
- Soil moisture monitoring systems use WSNs to monitor soil moisture at various locations.
- Surveillance systems use WSNs for collecting surveillance data (such as motion detection data)
- Smart grids use WSNs for monitoring the grid at various points.



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Examples of Wireless Sensor Networks - WSN >



Basics of ZigBee

- ZigBee is developed by ZigBee Alliance, ZigBee is a competitor of 6LOWPAN.
- ZigBee uses IEEE 802.15.4 at MAC and Physical Layer.
- ZigBee is designed for low-cost and low-power wireless IoT Networks.
- ZigBee is used in low data rate applications that requires long battery life and secure networking.
- ZigBee is simpler and less expensive than other WPANs such as Bluetooth and Wi-Fi.
- ZigBee operates at short range around 10m to 20m, networking range can be extended up to 500m.
- ZigBee operates at 2.4GHz unlicensed band.
- ZigBee operates at max data rate of 250 kbps.



Protocol Stack of ZigBee

Application Layer

ZigBee

Network Layer (Routing/ Network)

ZigBee

Data Link Layer (MAC – CSMA/CA)

IEEE 802.15.4

Physical Layer (2.4GHz, 250kbps)

IEEE 802.15.4

■ ZigBee supports different network topologies like:

- STAR
- Peer to Peer (Mesh)
- Cluster Tree

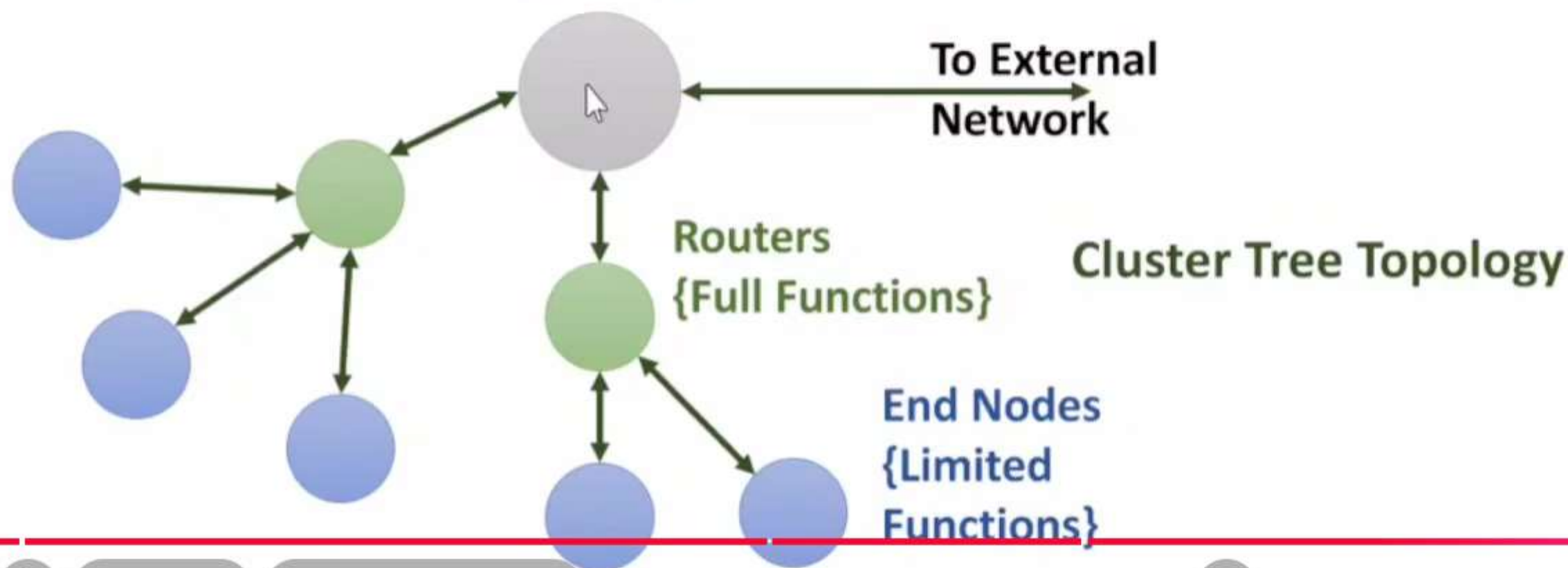
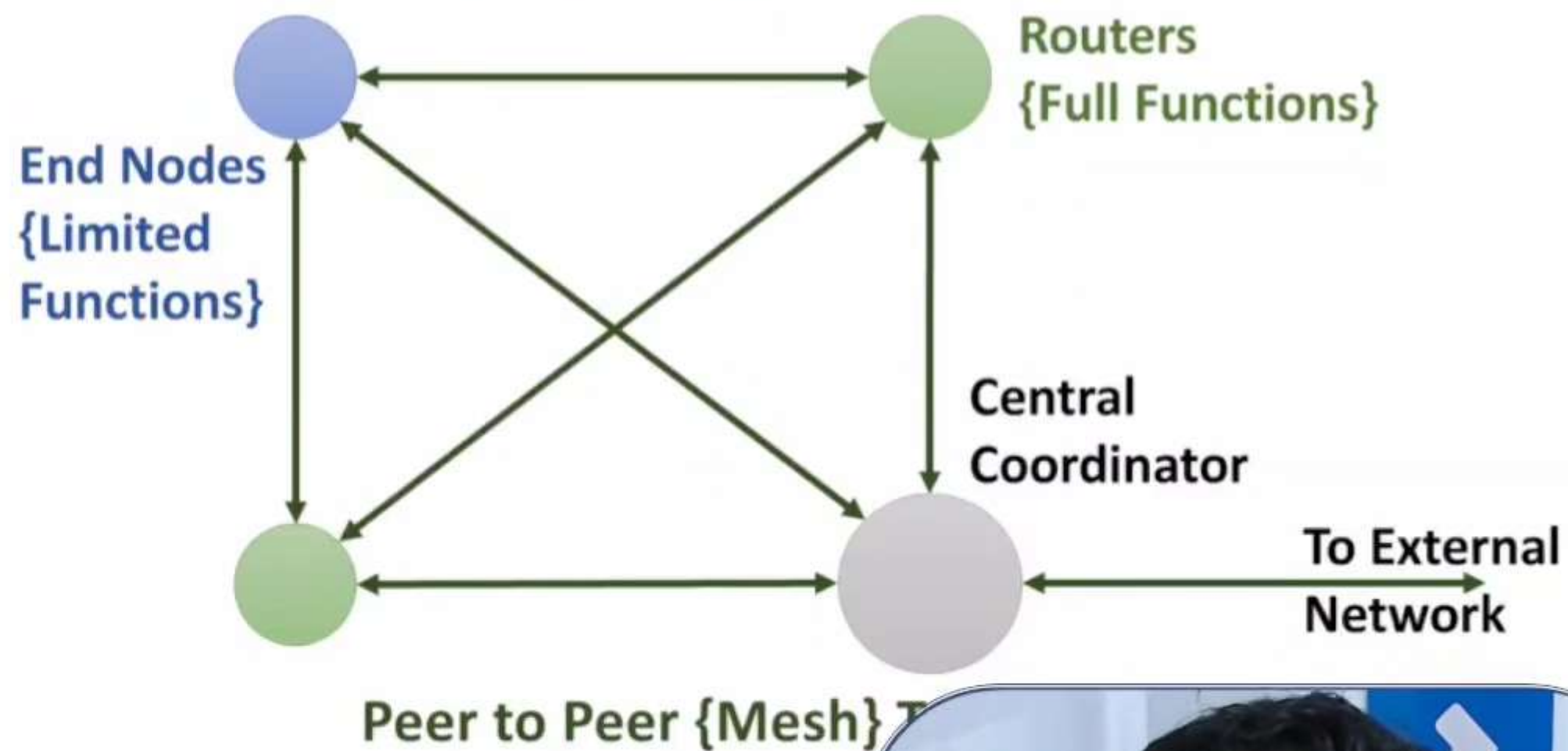
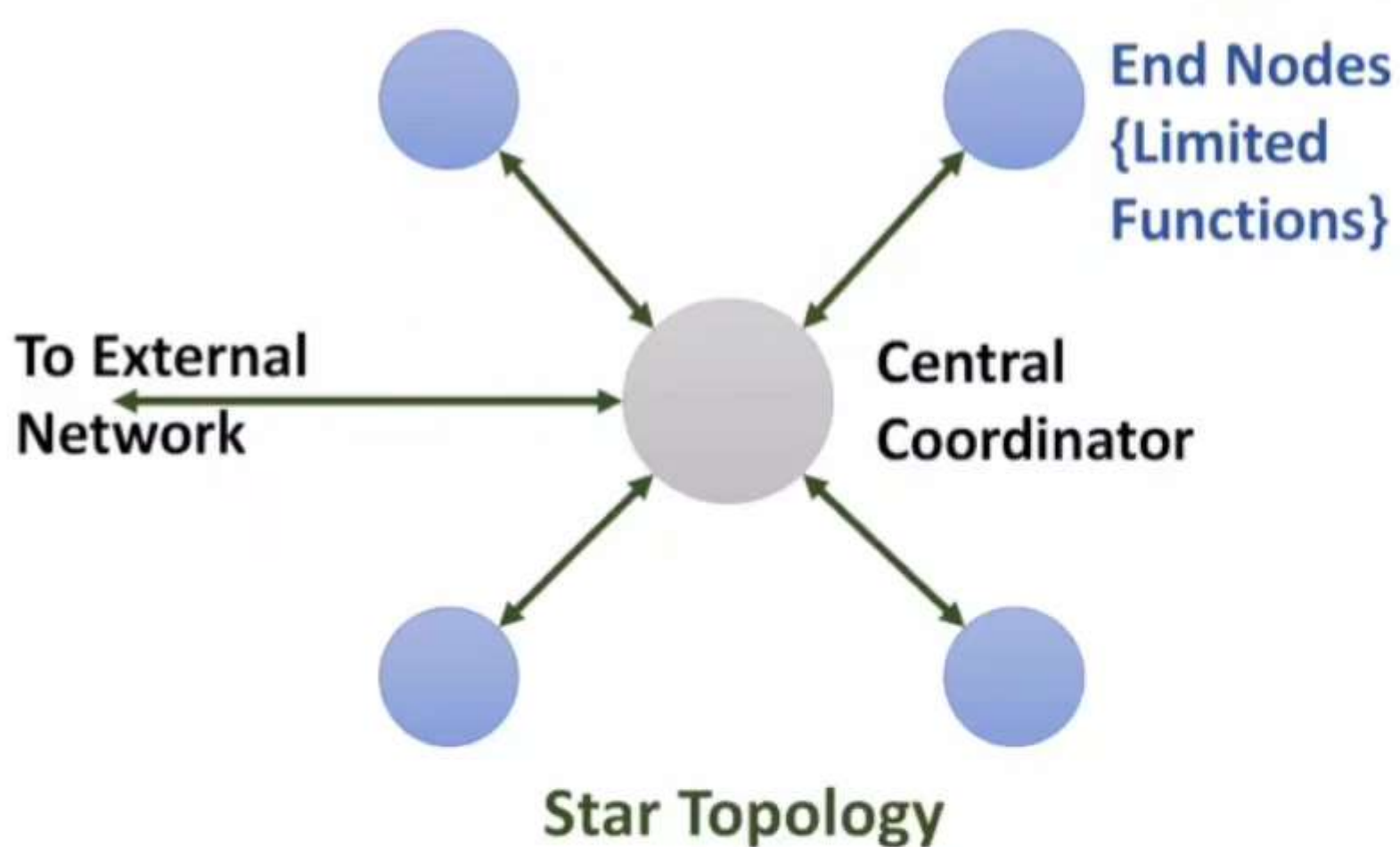
■ ZigBee can be specified as three different devices:

- ZigBee Coordinator
- ZigBee Router (Full Function Device)

ZigBee End Device (Reduced Function Device)



Topologies used in ZigBee



ZigBee Devices

■ ZigBee Coordinator

- In ZigBee Network, it has one coordinator node, that can store information.
- This Node can control a complete network, that can communicate with an external network.
- ZCs and ZRs require high power, so it can not be battery-powered.

■ ZigBee Router {Full Function Device}

- In ZigBee Network, it can route the data with interconnected nodes.
- It needs less memory than ZCs.
- It can also act as a coordinator in the network.

■ ZigBee End Device {Reduced Function Device}

- In ZigBee Network, it is the least expensive node with minimum
- It has a minimum power requirement, so it can operate with the
- It can only communicate with connected ZR or ZC.
- It communicates periodically for effective battery utilization.



ZigBee Advantages

- Easy to install and implement.
- It has low cost, low power, and long battery life.
- It supports many nodes (around 6500).
- It is more reliable and self-healing.



ZigBee Disadvantages

- It has a low data rate. {250kbps MAX}
- It is not as secure as Wi-Fi and Bluetooth.
- It requires additional devices ZCs and ZRs, which increases the cost.
- It lacks internet protocol support.
- ZigBee Networks are incompatible with other networks.

